Response to Your Second Review

GRU sensitivity, the LSTM comparison, and two bounded architecture checks

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Prepared 16 October 2026

Thank you for the detailed reply. This summarises what was actually done in response to each point, and what it found.

1. GRU Sensitivity to Optimiser and Learning Rate

Checked directly with a bounded sensitivity sweep, the same design already used for the GRU's hidden-size and sequence-length check. At one and two days ahead the default setting, Adam at a learning rate of 0.01, is at or near the best configuration tried. At three days ahead it is not: switching to SGD with momentum improves the mean error by about two points, with a tighter spread across seeds as well. This is the third separate sensitivity check, after architecture and Fourier harmonics, to flag three days ahead specifically as the GRU's weakest-tuned configuration, now stated together in the report's Limitations.

One part of the question does not apply to this architecture: a GRU's gate activations are fixed by the architecture itself, sigmoid and tanh, not a tunable choice the way a plain feed-forward layer's activation is, so “activation function” was not a free parameter to sweep.

2. The LSTM Comparison

This is the result that matters most. An LSTM was built and fitted under the exact same protocol as the GRU, the same channels, the same window, the same training setup, the same ten random seeds, and carried through the full rolling-origin evaluation rather than a bounded check.

The LSTM beats the GRU at every horizon: by 0.005 of relative error at one day ahead, 0.008 at two, and 0.013 at three, despite carrying roughly a third more parameters, 3,097 against 2,329. That result tests directly, and does not support, the reasoning the report originally gave for choosing the smaller architecture, that fewer parameters should matter more on a series this size. The report's Literature Review has been corrected to say so plainly rather than leave the original reasoning standing.

Against ARIMA, the LSTM's own picture closely mirrors the GRU's: a clear win at one day ahead (ahead in all ten seeds, reaching formal significance in five of them), a near-tie at two days, and a narrow ARIMA win at three days, with no seed reaching significance at either of the longer horizons.

3. Two Further Architectures, as Bounded Checks

Both were run as a single-split, three-seed check, the same bounded design already used elsewhere in this report for a first look at a question, rather than the full ten-seed treatment given to the GRU

and the LSTM. Building either one to that full standard would be a comparable amount of work to the LSTM's own treatment.

A small transformer encoder.

Your suggestion was conditional on data size, a transformer if the data is large; this project's data, about three thousand daily observations, is not, and the report's own Literature Review argues against a transformer on that basis. On the bounded check it does not beat ARIMA at any horizon, consistent with that argument. It is, however, more competitive with the GRU on this same comparison than the argument predicts, ahead of it at one and three days specifically.

A Kolmogorov-Arnold Network, as a KAN-LSTM hybrid.

Built exactly as you described it, an LSTM backbone with its usual linear output layer replaced by a Kolmogorov-Arnold layer. No maintained library exists for this the way one does for a GRU or LSTM, so the layer was implemented from the published construction and checked in isolation, confirming it can learn a nonlinear target, before being fitted on real data at all. It trained stably across all three seeds, with no divergence, which was a real risk going in. On the bounded check it is competitive with the other architectures without a dramatic win or loss in either direction, the worst of the four at one day ahead, the best at two.

What Changed in the Report

A new Methodology subsection gives the LSTM's full architecture and training details; its results sit alongside the GRU's throughout the Results chapter and its own row in the significance table. The GRU optimiser finding, the LSTM's results, and both bounded checks are stated in Limitations, and Future Work now includes completing the LSTM's own weather and threshold-sweep treatment and extending the two bounded checks to the full ten-seed evaluation. The report now compiles to 70 pages.